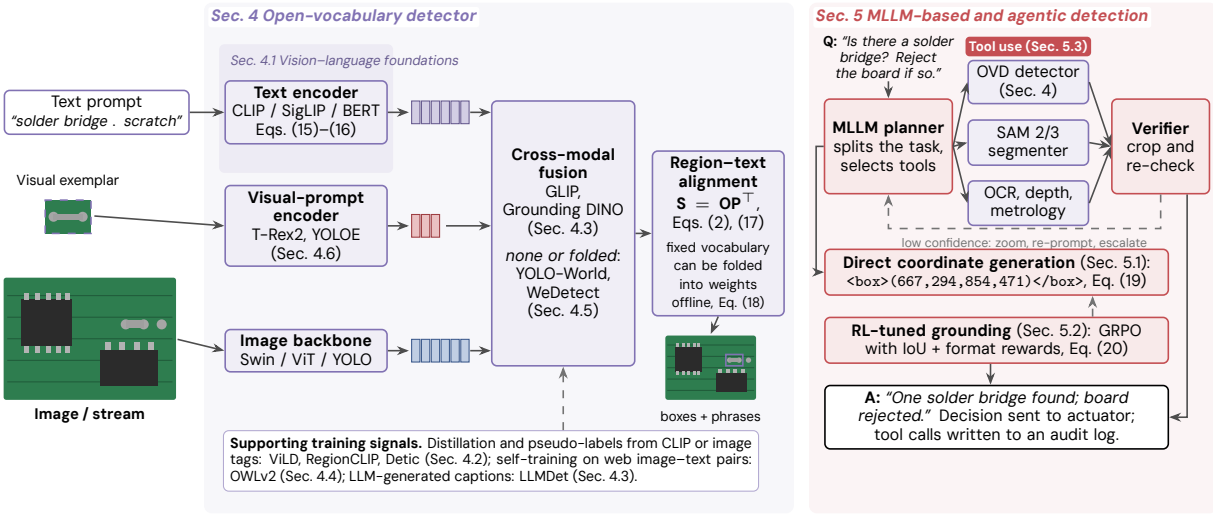
Sec. 4 Open-vocabulary detector
Sec. 4.1 Vision–language foundations
Text prompt
“solder bridge . scratch”
Text encoder
CLIP / SigLIP / BERT
Eqs. (15)–(16)
Visual exemplar
Visual-prompt encoder
T-Rex2, YOLOE
(Sec. 4.6)
Image backbone
Swin / ViT / YOLO
Image / stream
Cross-modal fusion
GLIP,
Grounding DINO
(Sec. 4.3)
none or folded:
YOLO-World,
WeDetect
(Sec. 4.5)
Region–text alignment
S = OP⊤,
Eqs. (2), (17)
fixed vocabulary can be folded into weights offline, Eq. (18)
boxes + phrases
Supporting training signals. Distillation and pseudo-labels from CLIP or image tags: ViLD, RegionCLIP, Detic (Sec. 4.2); self-training on web image–text pairs: OWLv2 (Sec. 4.4); LLM-generated captions: LLMDet (Sec. 4.3).
Sec. 5 MLLM-based and agentic detection
Q: “Is there a solder bridge? Reject the board if so.”
Tool use (Sec. 5.3)
MLLM planner
splits the task,
selects tools
OVD detector
(Sec. 4)
SAM 2/3
segmenter
OCR, depth,
metrology
Verifier
crop and
re-check
low confidence: zoom, re-prompt, escalate
Direct coordinate generation (Sec. 5.1):
<box>(667,294,854,471)</box>, Eq. (19)
RL-tuned grounding (Sec. 5.2): GRPO
with IoU + format rewards, Eq. (20)
A: “One solder bridge found; board rejected.” Decision sent to actuator;
tool calls written to an audit log.